# SEÁN Ó MURCHÚ

## Senior Software Engineer

sean@example.com +353 1 555 0100 Tallaght, Dublin, IE https://seanomurchu.dev seanomurchu
seanomurchu seanomurchu @seanomurchu @sean@fosstodon.org @seanomurchu
seanomurchu @seanomurchu talk recording

Backend engineer with eight years of experience designing distributed, event-driven systems. Specialises in functional programming, observability, and developer experience.

## EXPERIENCE

### Senior Software Engineer

**Acme Corp**

Jun 2022 – Present  Dublin, Ireland

*Platform team lead. Owns the event-sourcing stack underpinning the wider product surface.*

- Led the migration of a customer-facing monolith into a set of event-driven microservices, halving p99 latency.
- Designed and rolled out an event-sourcing platform now used by four product teams.
- Mentored three engineers from mid to senior level.

Scala Go Kafka Kubernetes

### Software Engineer

**Liffey Labs**

Jun 2019 – Jun 2021  Remote

- Built and shipped the first version of a SaaS product from scratch alongside a two-person team.
- Introduced the CI/CD pipeline and developer-onboarding flow that scaled the engineering org from 3 to 15.

Python Go AWS GitHub Actions

## VOLUNTEER

### Mentor

**CoderDojo Dublin**

Jun 2020 – Present

*Volunteer-led coding club introducing young people to programming.*

- Weekly mentoring sessions for 10–14 year-olds learning to code.
- Curate the Scala / functional-programming track.

Scala

## AREAS OF FOCUS

- Distributed systems: Kafka, CDC, idempotent consumers, event sourcing.
- Functional programming and type-driven design in Scala and Haskell.
- Developer experience: tooling, CI/CD, observability.
- Engineering culture: ADRs, RFCs, mentoring.

## SKILLS

**Languages** - Scala Haskell Python Go Rust

**Infra** - Kafka AWS Terraform Docker Kubernetes

## LANGUAGES

English ●●●●●○

Irish ●●●○○○

French ●●●●○○

Portuguese ●●◐○○○

## EDUCATION

### M.Sc. in Computer Science

**Tallaght Institute of Technology**

2017 – 2019

Distinction Distributed Systems Type Theory Concurrency

### Computer Science

**Trinity College Dublin**

2014 – 2017

# PROJECTS

### Application

*Hyperion*

Acme Corp

*Distributed task scheduler in Rust*

**Lead** **Architect**

Jan 2024 – Aug 2024

- Handled 10k tasks/s on a single node.
- Designed an idempotent retry protocol.

Rust Tokio gRPC

*Crucible*

*Migration tool for legacy databases*

**Maintainer**

Sep 2024 – Present

- Schema diffing across PostgreSQL and MySQL.

### Library

*Tinkerbell*

*Tiny IRC bot*

Lua

### Other

*Quill*

# PUBLICATIONS

### Articles

- *Event Sourcing in Practice*.
  Personal Blog
  14 Jun 2024
  A walk-through of idempotent Kafka consumers and the replay strategy that ships with them.
- *Untyped Note*.
  Jan 2024

### Conference Papers

- *Idempotent Kafka Consumers at Scale*.
  EuroSys
  Apr 2024

### Talks

- *Designing for Failure*.
  ScalaIO
  Sep 2023

### Books

- *Functional Domain Modelling*.
  Self-published
  2022

# INTERESTS

**Music** - Trad Jazz

**Sport** - Hurling Climbing

**Travel** - Japan Iceland

# CERTIFICATIONS

CNCF

Certified Kubernetes Administrator · Aug 2023

Certified Kubernetes Application Developer · Jan 2024

AWS

AWS Solutions Architect — Professional · May 2023

AWS DevOps Engineer — Professional · Sep 2024

Hashicorp Terraform Associate · Dec 2022

Certified Kubernetes Security Specialist · Mar 2024

Google Cloud Professional Architect · Nov 2023

# AWARDS

*Best Paper Award*

**ScalaConf**

Sep 2024

For *Idempotent Kafka Consumers*.

*Dean's List*

# REFERENCES

Dr Ada Lovelace

*Seán is a calm, methodical engineer who left every system he touched in a better state than he found it. Recommended without reservation.*

*Pragmatic, curious, unfailingly kind — would hire again in a heartbeat.*